

Jonathan Luna

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Electrical Engineering, Minor in Mathematics

Aug. 2023 – April 2026

- GPA: 3.5 (Dean's List, Spring 2025)
- **Planned Graduate Studies:** M.S. in Electrical Engineering (Enrollment: Fall 2026)
- **Relevant Coursework:** Digital Control Systems, Intro to Radar Systems, Semiconductor Material & Device Characterization, Electronics 2, Real Time Systems, AI/ML Foundations
- **Involvement:** Institute of Electrical and Electronics Engineers (IEEE)

PROJECTS

Radar Design Simulation | *MATLAB, Signal Processing*

Feb. 2025 – April 2025

- Designed a 150 GHz airborne radar achieving 100% mandatory spec compliance (16/16) and a 77.59% evaluation score, placing 3rd against graduate teams as a solo competitor.
- Conducted system-level trade studies across PRF, transmit power, antenna dimensions, and spin rate to simultaneously satisfy all 16 hard requirements while optimizing stretch goals, achieving $P_d \geq 0.91$ across near, boresight, and far range at a 0.80 threshold.
- Optimized swath coverage to 50.33 km ($5\times$ the 10 km minimum, meeting the 50 km stretch goal) while maintaining a boresight grazing angle of 5.4° and 10 scans on target for robust multi-look detection.
- Balanced competing constraints between transmit power, duty cycle, and PRF to sustain Nyquist-valid Doppler measurement at 160 kHz against a max Doppler of ~ 157.6 kHz, within a 700 K system noise temperature.

Smart Multi-Port USB Charging System | *FreeRTOS, PCB Design, Power Electronics*

Oct. 2025 – Present

- Designed and fabricated a 2-layer, 4-port USB charging system delivering up to 2A per port at 5V, completing a first hardware iteration with functional validation and a second iteration currently in layout for corrections and improvements.
- Performed trace width analysis across all high-current power paths to ensure current capacity at 2A per port, selecting trace geometries to minimize resistive drop and thermal rise along the 5V distribution network.
- Constructed a mixed-voltage power system integrating a 5V rail for TPS22918 load switches across 4 independent charging channels and a 3.3V rail for MCU operation, with LM2670 and MP1584 buck converters handling regulation.
- Validated system behavior under load using an electronic load across expected charging conditions, identifying layout issues that drove the second PCB revision.

Ultrasonic Sensor | *Embedded C, MSP430, PCB Design*

June 2025 – Aug. 2025

- Designed and assembled a mixed through-hole and SMD PCB integrating an MSP430 MCU and onboard power regulation, achieving ± 2 mm distance measurement accuracy over a 100 cm range.
- Implemented a multi-output feedback system in embedded C, simultaneously driving a PWM-controlled LED indicator, LCD distance display, and UART serial output for real-time monitoring.
- Validated end-to-end system performance through hardware bring-up and debugging, confirming sensor integration, power rail stability, and firmware timing across all output interfaces.

EXPERIENCE

Undergraduate Learning Assistant (Linear Circuits I & II)

Aug. 2024 – Present

University of Central Florida

Orlando, FL

- Leading 4 weekly sessions for 180+ students, resulting in a **10% average exam score increase**.
- Developed MATLAB-based Laplace Transform visualizations to bridge student learning gaps.
- Providing targeted one-on-one guidance during office hours for 50+ students weekly.

Undergraduate Teaching Assistant (Linear Control Systems)

May 2025 – Aug. 2025

University of Central Florida

Orlando, FL

- Created MATLAB scripts to automate detection of intersections between Root Locus and Damping Ratio lines.
- Managed weekly grading for 45+ students, providing feedback that boosted performance by 5%.

SKILLS SUMMARY

Software: LTSpice, MATLAB, Autodesk Fusion (PCB Design & CAD), KiCad, C/C++, Python, FreeRTOS

Hardware: Oscilloscope, Function Generator, Multimeter, Soldering, MSP430, ESP32

Languages: English (Native), Spanish (Intermediate), French (Intermediate)